



الجامعة الإسلامية العالمية
International Islamic University, Islamabad

Retail Store Sales Data Analysis and Predictive Modeling Using RapidMiner

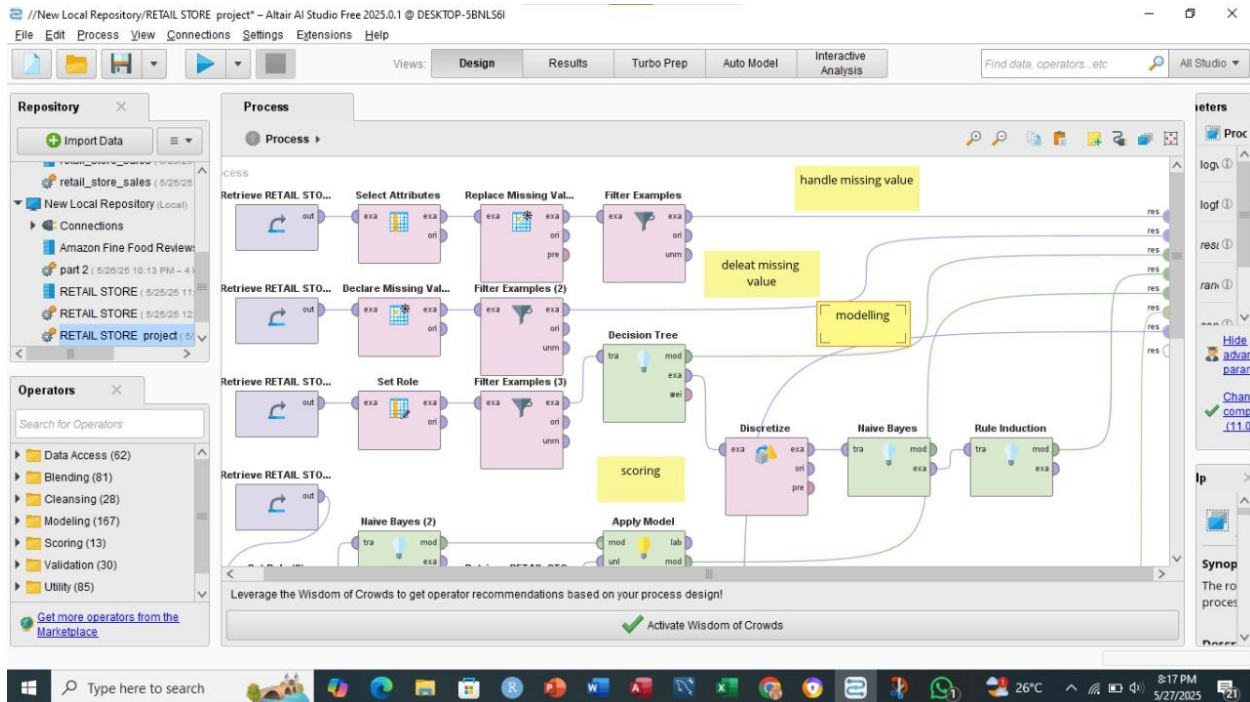
ADEENA YASEEN 08-FMS/BSBA/F21

SUBMITTED TO: Dr.SARA QAISER

SUBJECT : DATA MINNING AND WAREHOUSING

1. Project Title

Retail Store Sales Data Analysis and Predictive Modeling Using RapidMiner



2. Executive Summary

This project explores retail transaction data using RapidMiner to derive actionable insights. The dataset contains structured fields such as Transaction ID, Customer ID, Category, Item, Price Per Unit, Quantity, Total Spent, Payment Method, Location, Transaction Date, and Discount Applied.

We applied data preprocessing, classification models (Decision Tree, Naive Bayes, Rule Induction), validation techniques, and interpretation methods to support business decision-making..

3. Introduction

Retail data provides valuable insights into customer behavior, product popularity, and purchasing trends. However, raw data must be cleaned and processed before it becomes meaningful. This

project uses RapidMiner to perform a complete data science pipeline on retail store sales data, which includes structured and unstructured (text) data.

The insights obtained from this analysis can help in understanding customer segments, improving inventory management, and enhancing sales performance.

4. Tools and Technologies Used

- **Software Platform:** RapidMiner Studio (Version 2025.0.1)
 - **Libraries Installed:**
 - **MeaningCloud:** Used for sentiment analysis and language processing
 - **Operator Toolbox:** Provides advanced operators for custom workflows
 - **Text Processing:** Allows tokenization, stemming, filtering, and transformation of text data
-

5. Dataset Description

Dataset Columns:

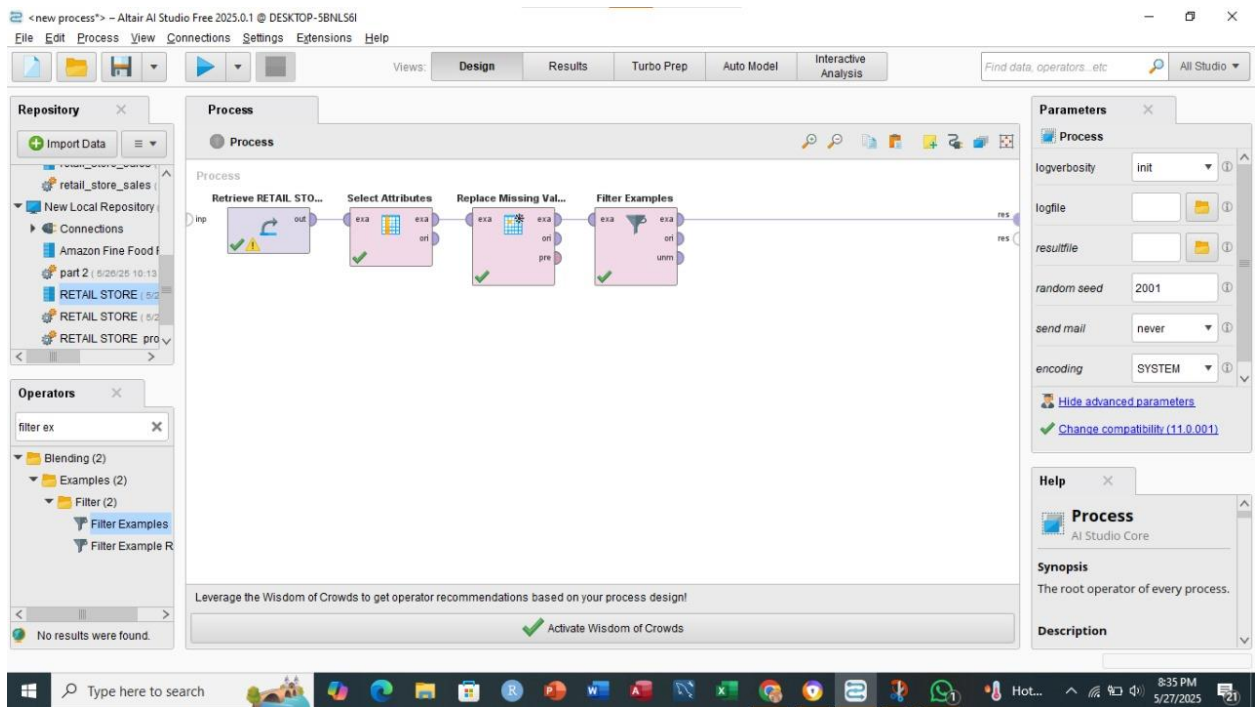
- Transaction ID – Unique transaction reference
 - Customer ID – Identifier for individual customers
 - Category – Product category (e.g., Electronics, Grocery)
 - Item – Specific product name
 - Price Per Unit – Cost per item
 - Quantity – Number of units purchased
 - Total Spent – Price * Quantity
 - Payment Method – Mode of payment (e.g., Cash, Credit Card)
 - Location – Store location
 - Transaction Date – Date of purchase
 - Discount Applied – Binary field (Yes/No) indicating discounts
-

6. Data Preprocessing

6.1 Handling Missing Values

1. **Retrieve RETAIL STORE SALES:** Loads the dataset from the repository.
2. **Select Attributes:** Keeps only the relevant columns needed for analysis.
3. **Replace Missing Values:** Fills in missing data (e.g., with mean or mode) to ensure completeness.
4. **Filter Examples:** Removes invalid rows (e.g., zero quantity or missing fields) to clean the data.

This preprocessing workflow prepares the dataset for accurate analysis and modeling.



Result:

The screenshot displays the Altair AI Studio interface. The main window shows a data table with 14 rows and 10 columns. A yellow warning banner at the top of the table states: "The process is only using 10,000 rows of data, which may reduce model accuracy up to 30%. Upgrade your license to use all your data! Learn more about license limits." The table data is as follows:

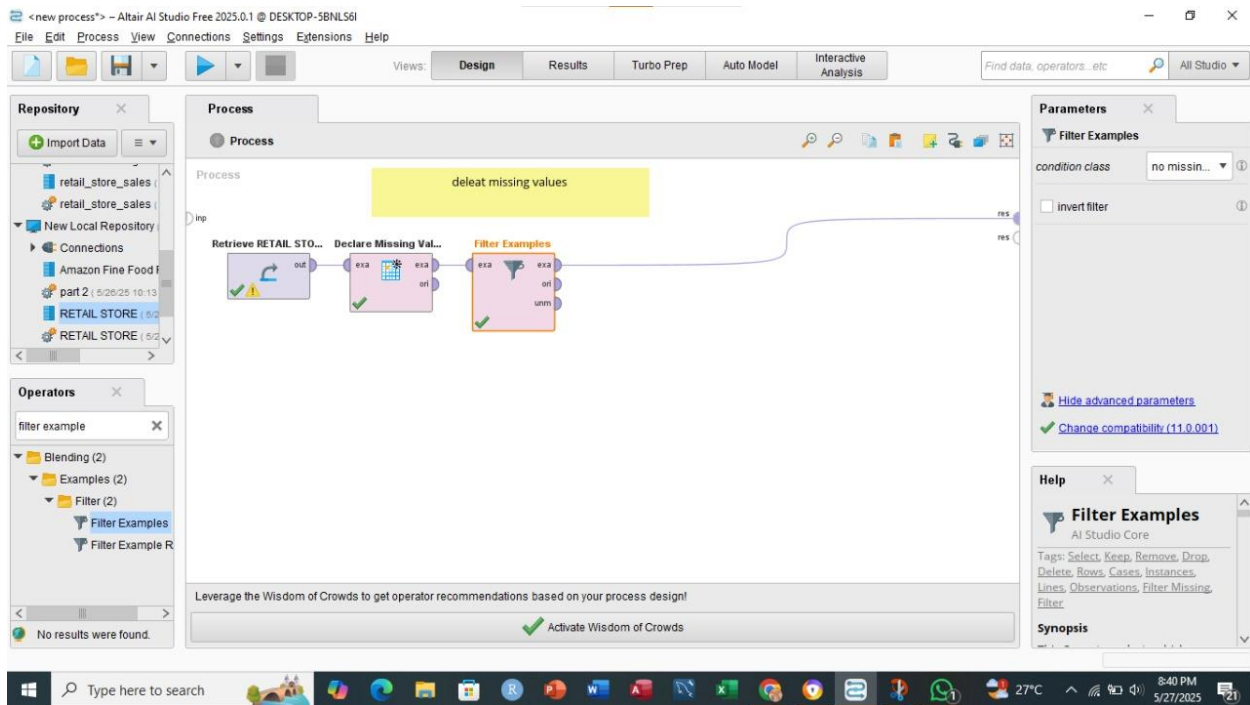
Row No.	Transaction ...	Item	Price Per Unit	Quantity	Total Spent	Payment Me...	Location	Discount Ap...	Tran
1	TXN_6867343	Item_20_BUT	23.257	6	127.287	Cash	Online	FALSE	?
2	TXN_6867343	Item_20_BUT	23.257	6	127.287	Cash	Online	FALSE	?
3	TXN_6867343	Item_20_BUT	23.257	6	127.287	Cash	Online	FALSE	?
4	TXN_6867343	Item_20_BUT	23.257	6	127.287	Cash	Online	FALSE	?
5	TXN_6867343	Item_20_BUT	23.257	6	127.287	Cash	Online	FALSE	?
6	TXN_6867343	Item_20_BUT	23.257	6	127.287	Cash	Online	FALSE	?
7	TXN_6867343	Item_20_BUT	23.257	6	127.287	Cash	Online	FALSE	?
8	TXN_6867343	Item_20_BUT	23.257	6	127.287	Cash	Online	FALSE	?
9	TXN_6867343	Item_20_BUT	23.257	6	127.287	Cash	Online	FALSE	?
10	TXN_6867343	Item_20_BUT	23.257	6	127.287	Cash	Online	FALSE	?
11	TXN_6867343	Item_20_BUT	23.257	6	127.287	Cash	Online	FALSE	?
12	TXN_6867343	Item_20_BUT	23.257	6	127.287	Cash	Online	FALSE	?
13	TXN_6867343	Item_20_BUT	23.257	6	127.287	Cash	Online	FALSE	?
14	TXN_6867343	Item_20_BUT	23.257	6	127.287	Cash	Online	FALSE	?

The right sidebar shows a "Repository" panel with a tree view of data sources. The "New Local Repository (Local)" section is expanded, showing "Connections" and "Processes". The "Connections" section includes "Amazon Fine Food Reviews", "part 2", "RETAIL STORE", and "RETAIL STORE project". The "Processes" section includes "Amazon Fine Food Reviews", "retail_store_sales", and "retail_store_sales".

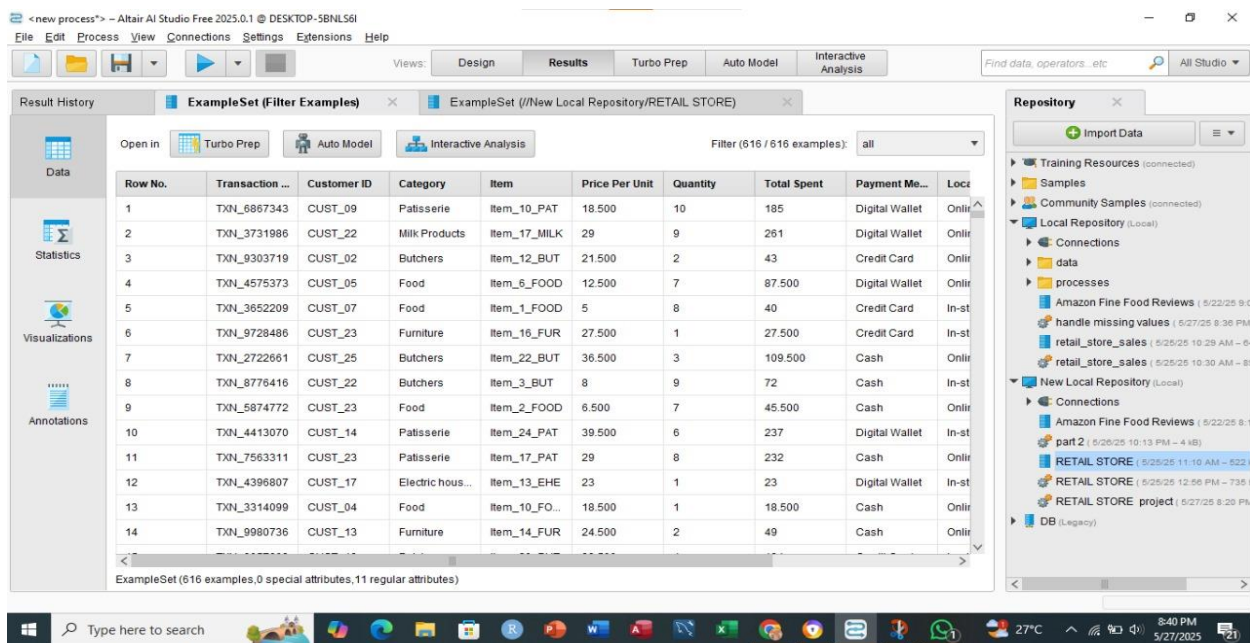
DELETE MISSING VALUES

1. **Retrieve RETAIL STORE SALES:** Imports the dataset from the repository.
2. **Declare Missing Values:** Identifies which values are missing in the dataset.
3. **Filter Examples:** Removes all rows (examples) that have any missing values using the condition no missing attributes.

✓ This process is used to **clean the dataset by deleting rows with missing values** to ensure data quality before analysis or modeling.



Result:

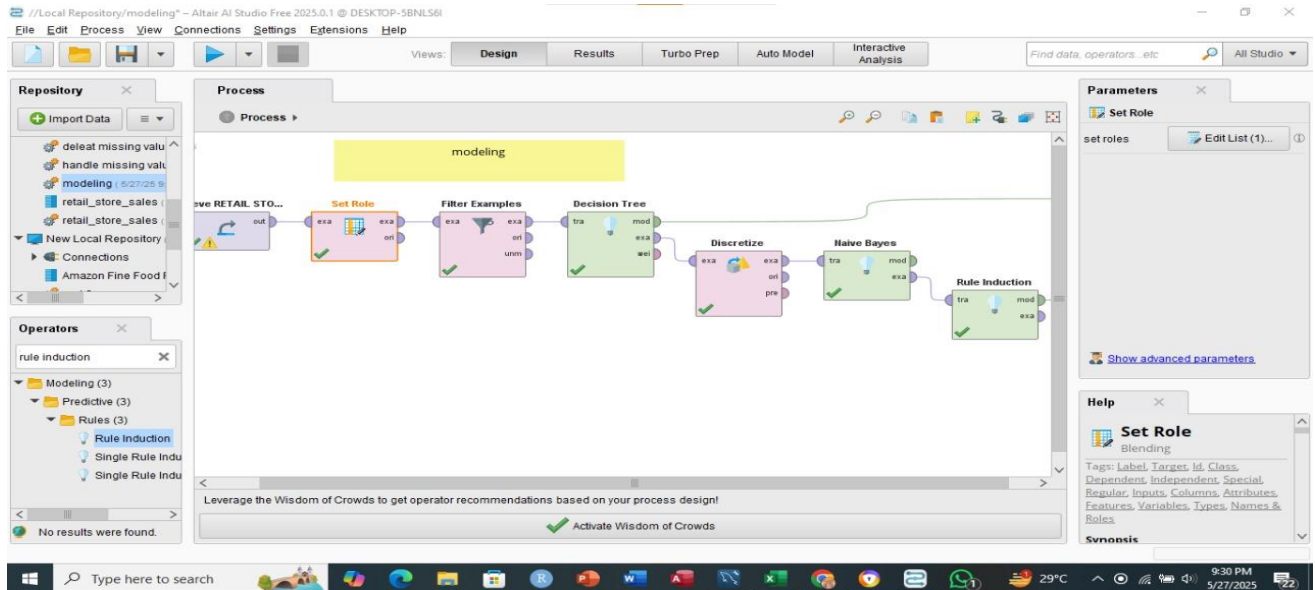


7. Data Modeling

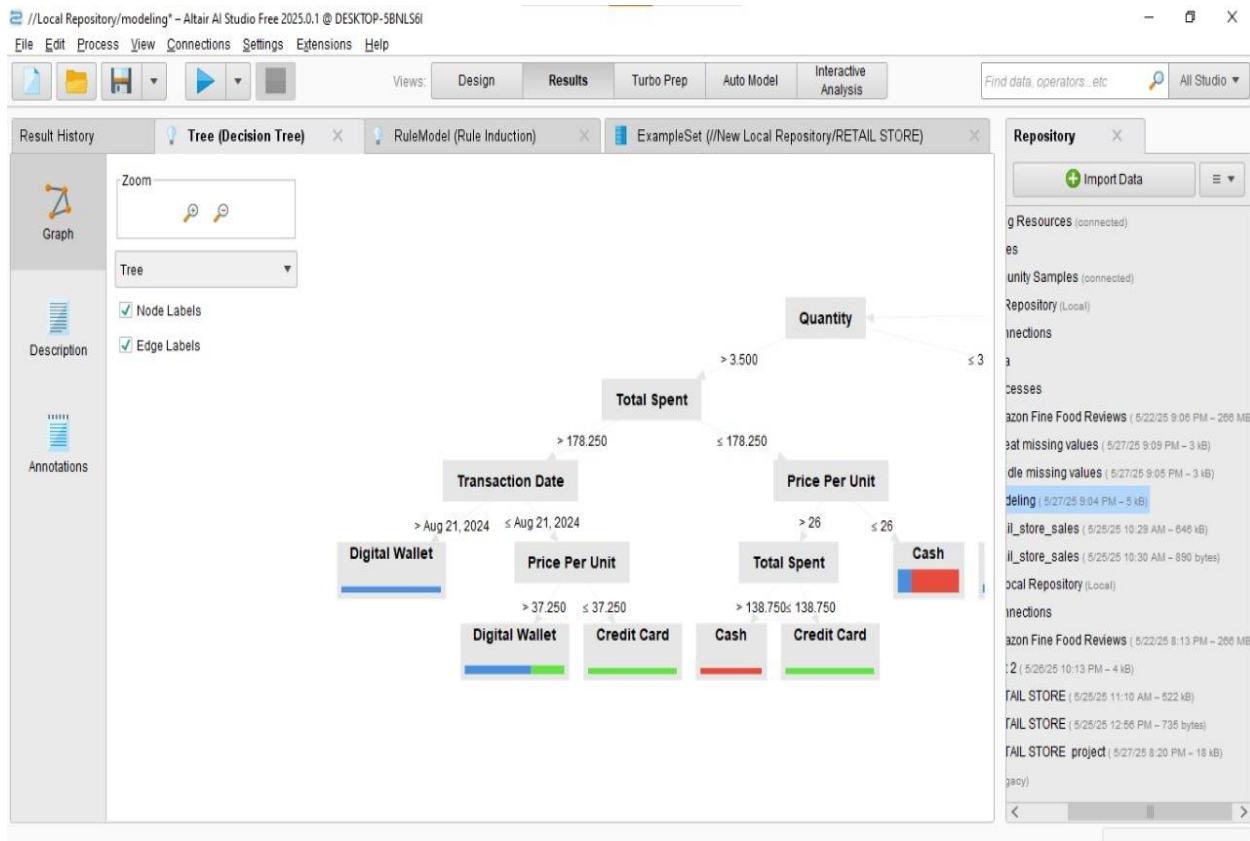
1. **Retrieve RETAIL STORE SALES:** Loads the dataset.

2. **Set Role:** Defines the target variable (label PAYMNT METHOD) and IDs.
3. **Filter Examples:** Removes invalid or irrelevant data rows.
4. **Discretize:** Converts continuous numeric data into categories (e.g., grouping Total Spent into "Low", "Medium", "High").
5. **Decision Tree:** Builds a tree-based model to classify or predict based on input features.
6. **Naive Bayes:** Applies a probabilistic model assuming attribute independence.
7. **Rule Induction:** Generates simple IF-THEN rules for easy interpretation of patterns.

✓ This setup trains **three classification models** to find patterns in the dataset for prediction or decision-making.



Results:



8. Scoring and Validation Scoring in RapidMiner using Naive Bayes

Scoring is the process of applying a trained machine learning model to new, unseen data in order to generate predictions. In our project, we utilized the **Naive Bayes** algorithm for classification and performed scoring on the **retail_store_sales** dataset using the following steps:

Workflow Description

1. Retrieve Training Data

The dataset `retail_store_sales` was imported and used to train the Naive Bayes model. The `Retrieve` operator fetched this dataset into the process.

2. Set Role

A `Set Role` operator was used to define the target variable (label) for prediction. In our case, the attribute indicating the **payment method** (Cash, Credit Card, Digital Wallet) was set as the label.

3. Train Naive Bayes Model

The `Naive Bayes` operator was applied to the dataset to train the model. This probabilistic classifier assumes independence among predictors and calculates the likelihood of each class based on input features.

4. Retrieve Unlabeled Data

A second `Retrieve` operator was used to bring in another set of customer records (presumably without the label) for scoring purposes.

5. Apply Model (Scoring)

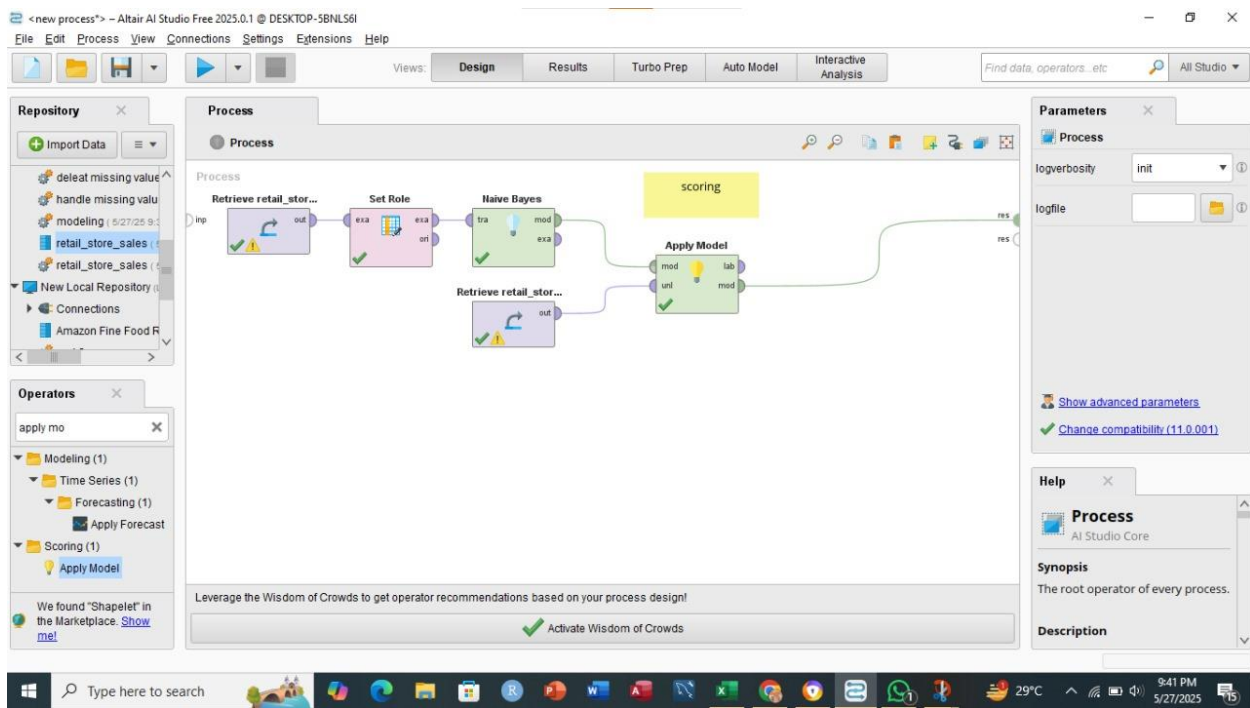
The `Apply Model` operator was used to apply the trained Naive Bayes model to the new unlabeled dataset. This generated predicted labels for each instance, indicating the likely payment method each customer would use.

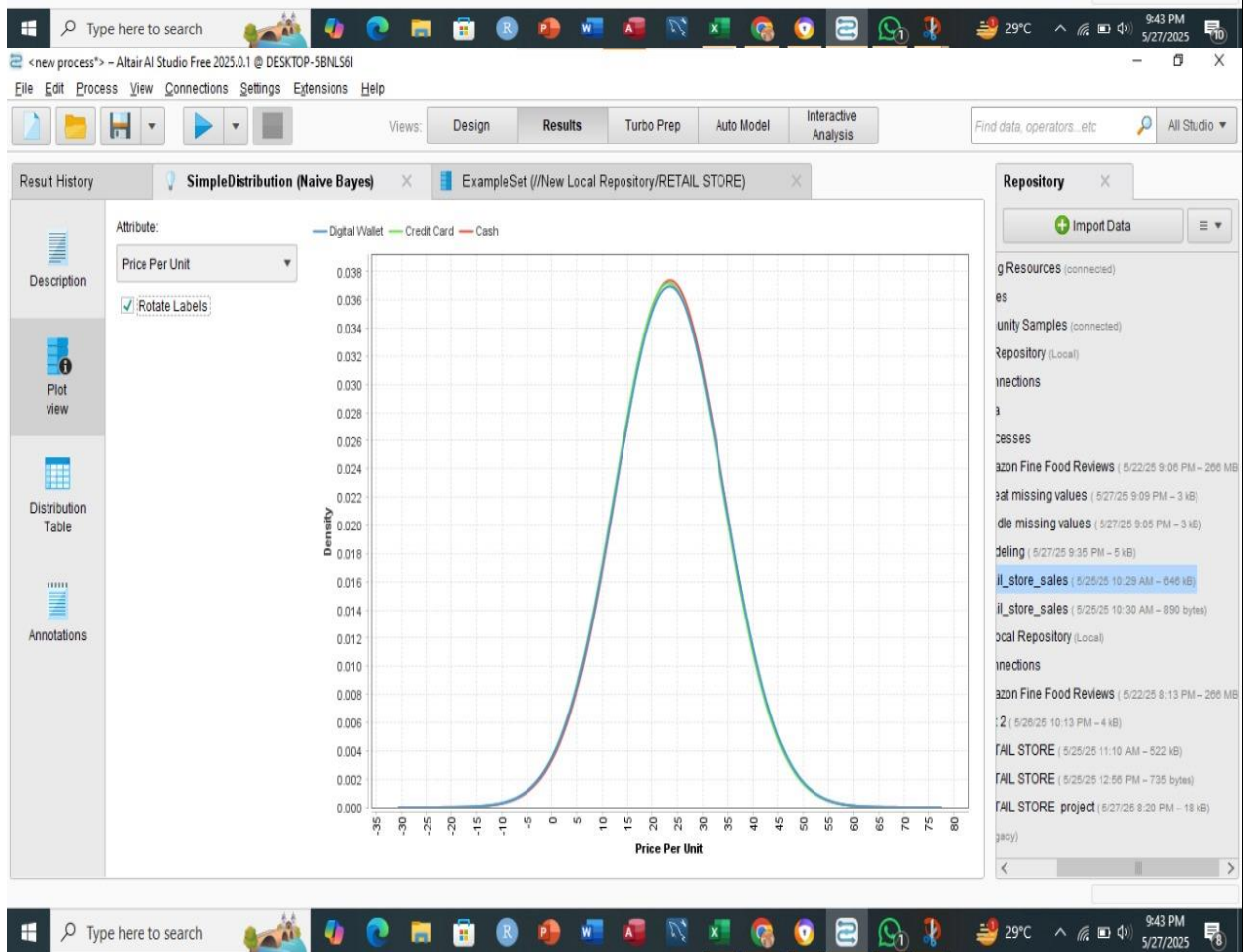
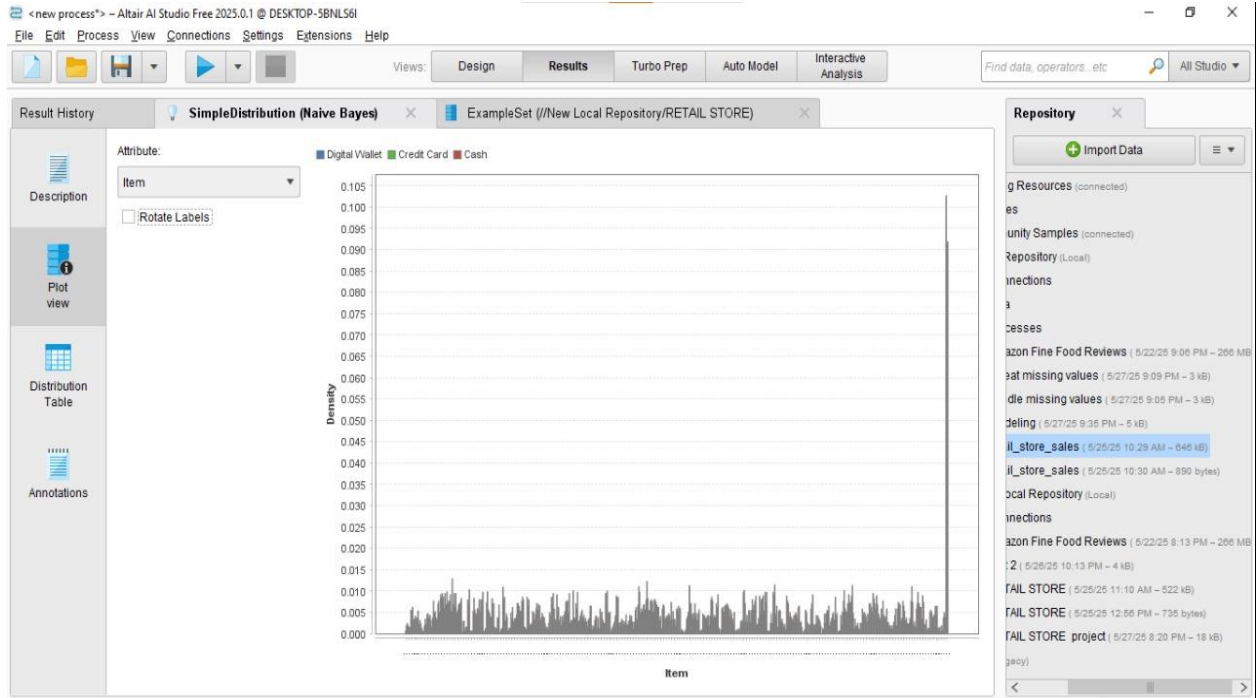
6. Results and Visualization

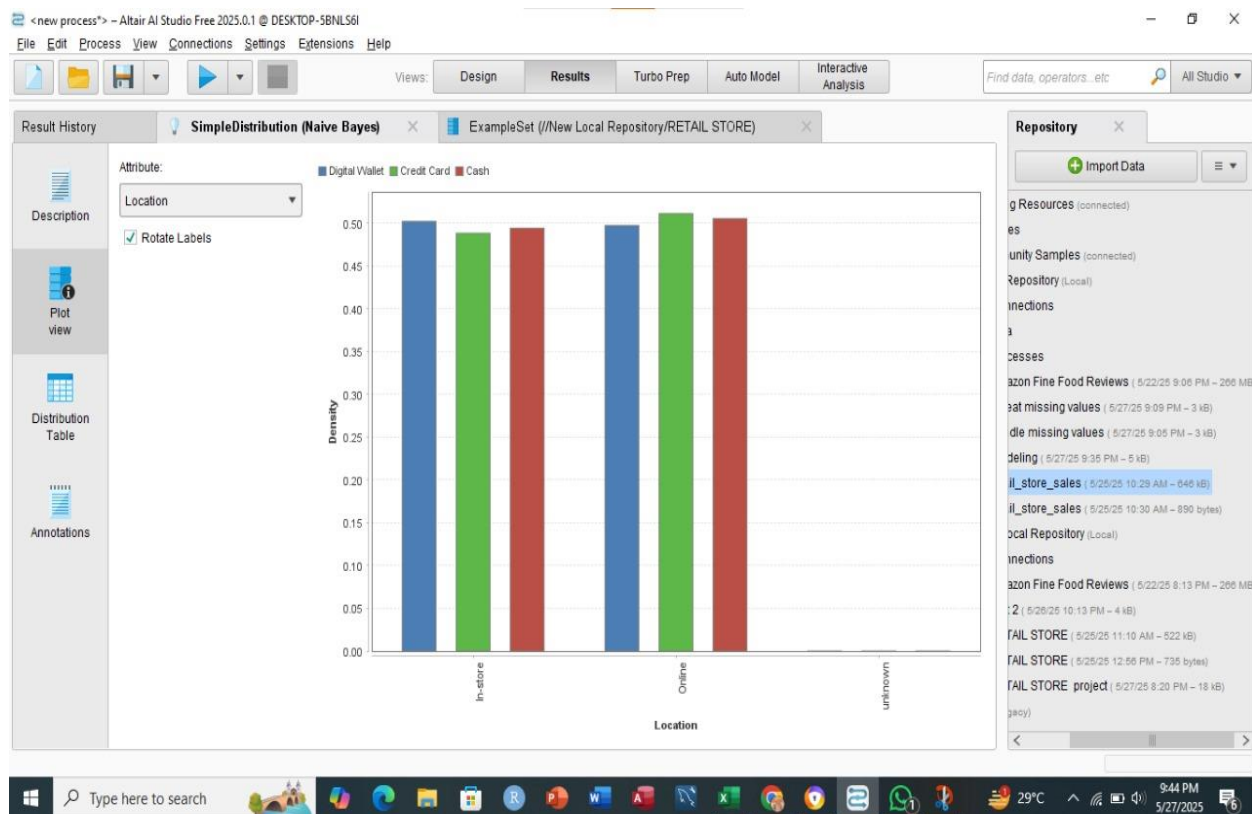
The results were visualized using a **bar chart**, which showed the **distribution of predicted payment methods (Cash, Credit Card, Digital Wallet)** across customers (CUST_01 to CUST_25). Each bar represented the **probability density** for each payment method.

Insights from Scoring Results

- The plot view illustrates that each customer has varying probabilities for each payment method.
- This scoring helps in identifying the **preferred mode of payment** per customer, which can guide retail decision-making such as targeted promotions or personalized recommendations.
- For instance, if a customer has a higher predicted probability for using digital wallets, they can be targeted with wallet-based offers or discounts.







Test Split and Validation

To evaluate the performance of our machine learning models, we used a technique called **train-test split**. This method ensures that our models are trained on one part of the data and tested on another unseen part, which gives us a better understanding of how the model will perform in the real world.

In RapidMiner, we used the "**Split Data**" operator to divide our dataset:

- **70% of the data was used for training** – this part was used to teach the model how to make predictions.
- **30% of the data was used for testing** – this part was used to evaluate how well the model can predict on new, unseen data.

We then applied the trained model to the test set using the "**Apply Model**" operator. After that, we used the "**Performance**" operator to calculate the model's evaluation metrics.

Altair AI Studio Free 2025.0.1 @ DESKTOP-5BNLS6I

File Edit Process View Connections Settings Extensions Help

Views: Design Results Turbo Prep Auto Model Interactive Analysis

Find data, operators, etc. All Studio

Repository

- Import Data
- retail_store_sales (5/27/25 9:47)
- retail_store_sales (5/27/25 9:47)
- scoring (5/27/25 9:47)
- test split (5/27/25 9:58)
- New Local Repository (5/26/25 10:13)
- Connections
- Amazon Fine Food Reviews (5/26/25 10:13)
- part 2 (5/26/25 10:13)

Operators

set role

- Blending (1)
- Attributes (1)
- Names & Roles (1)
- Set Role

No results were found.

Process

Process

test split and validation

Retrieve retail_store_sales (5/27/25 9:47) → Split Data (5/27/25 9:47) → Set Role (5/27/25 9:47) → Naive Bayes (5/27/25 9:47) → Apply Model (5/27/25 9:47) → Set Role (2) (5/27/25 9:47) → Performance (5/27/25 9:47)

Parameters

Process

logverbosity: init

logfile:

Show advanced parameters

Change compatibility (11.0.001)

Help

Process

AI Studio Core

Synopsis

The root operator of every process.

Description

Leverage the Wisdom of Crowds to get operator recommendations based on your process design!

Activate Wisdom of Crowds

RESULTS:

Altair AI Studio Free 2025.0.1 @ DESKTOP-5BNLS6I

File Edit Process View Connections Settings Extensions Help

Views: Design Results Turbo Prep Auto Model Interactive Analysis

Find data, operators, etc. All Studio

Result History

PerformanceVector (Performance) ExampleSet (/New Local Repository/RETAIL STORE)

Criterion

- accuracy
- precision
- recall
- AUC (optimistic)
- AUC
- AUC (pessimistic)

Description

Annotations

Table View Plot View

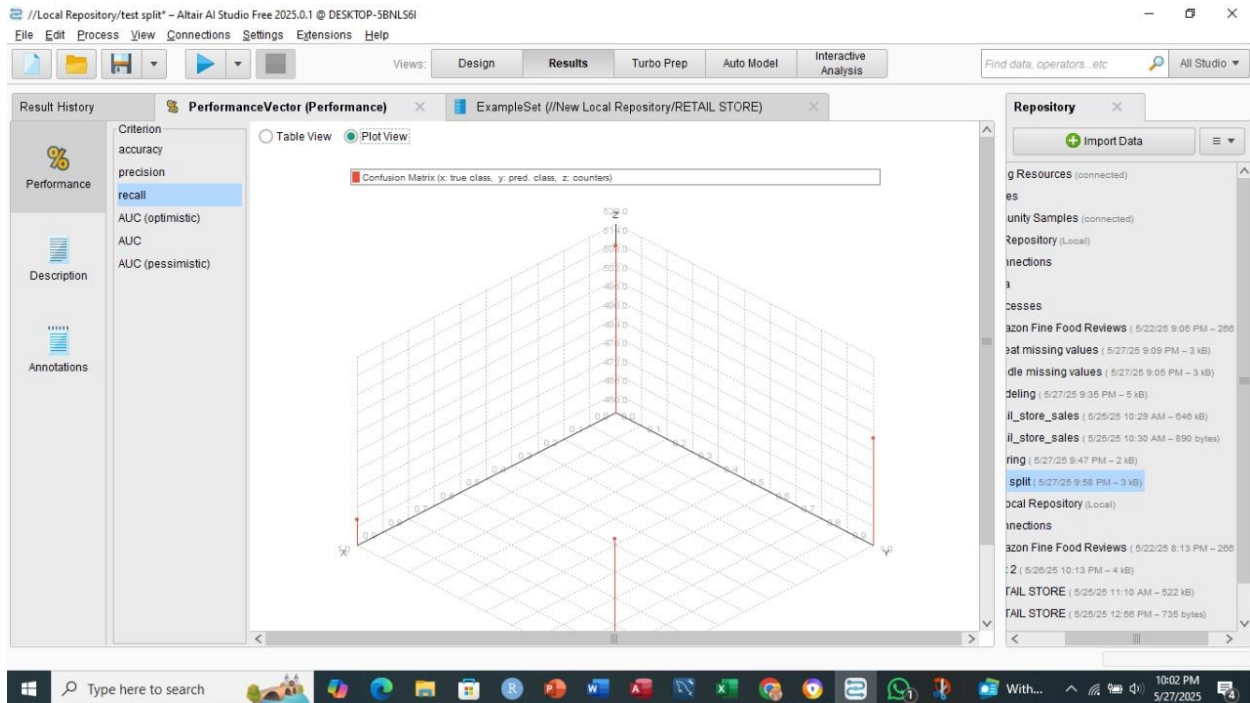
accuracy: 51.39%

	true True	true False	class precision
pred. True	513	468	52.29%
pred. False	494	504	50.50%
class recall	50.94%	51.85%	

Repository

Import Data

- g Resources (connected)
- es
- unity Samples (connected)
- Repository (Local)
- injections
- Classes
- amazon Fine Food Reviews (5/22/25 9:06 PM - 206)
- cat missing values (5/27/25 9:09 PM - 3 kB)
- die missing values (5/27/25 9:05 PM - 3 kB)
- deling (5/27/25 9:35 PM - 5 kB)
- il_store_sales (5/26/25 10:29 AM - 646 kB)
- il_store_sales (5/26/25 10:30 AM - 890 bytes)
- ring (5/27/25 9:47 PM - 2 kB)
- split (5/27/25 9:58 PM - 3 kB)
- ocal Repository (Local)
- injections
- amazon Fine Food Reviews (5/22/25 8:13 PM - 206)
- 2 (5/26/25 10:13 PM - 4 kB)
- FAIL STORE (5/26/25 11:10 AM - 522 kB)
- FAIL STORE (5/26/25 12:56 PM - 735 bytes)



Text Mining

9.1 Library Setup

We installed:

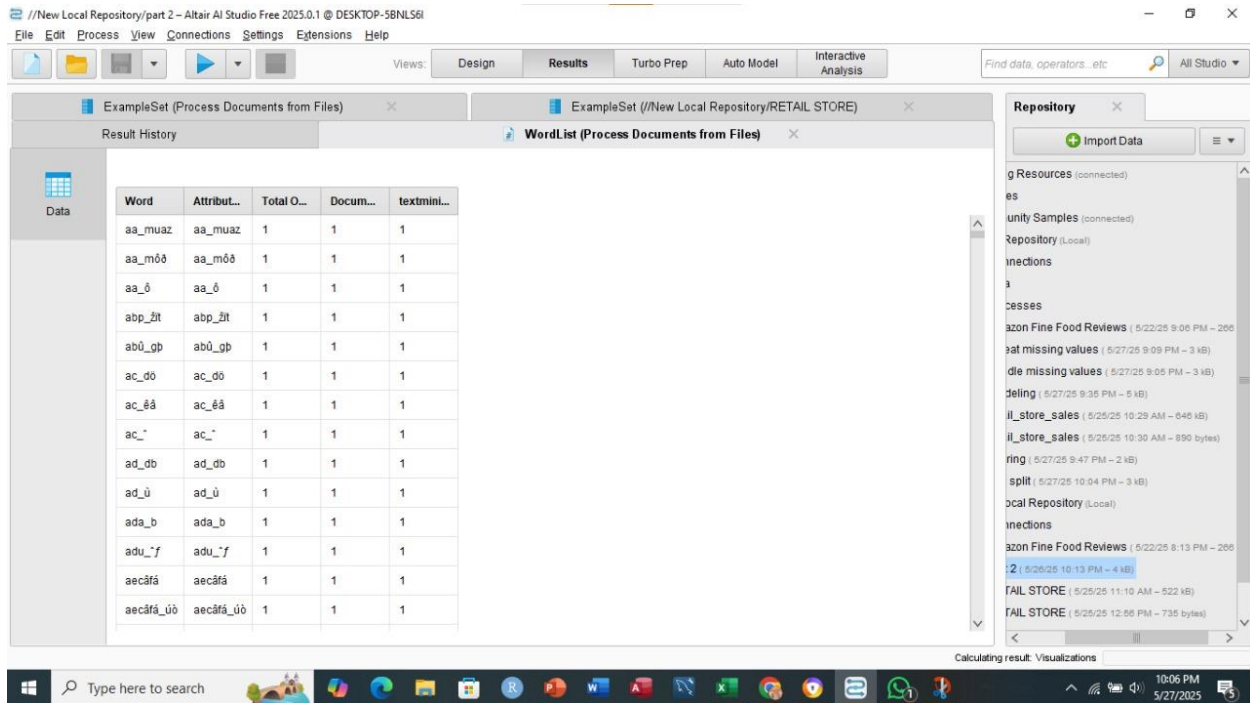
- **MeaningCloud** for sentiment analysis
- **Text Processing** for tokenization, filtering, stemming
- **Operator Toolbox** for extra functionalities

9.2 Text Mining Process

We processed product descriptions or review texts using:

- **Process Documents from Files**
- **Tokenization:** Split text into words
- **Filtering Stopwords:** Remove common words like "the", "is"
- **Stemming:** Convert words to root form (e.g., "buying" → "buy")

The texts were then converted into **word vectors** for analysis.



Results and Insights

- **Top Products:** Products with highest quantity sold and revenue
- **Customer Segments:** E.g., women aged 25-35 preferred fashion-related categories
- **Seasonal Trends:** Sales spikes seen during festive months
- **Text Analysis:** Sentiment scores were generally positive; keywords like “value”, “quality”, “fast” were frequent

These insights can help the retail store optimize inventory, target promotions, and improve customer satisfaction.

11. Conclusion

This project shows the complete pipeline of a retail analytics workflow using RapidMiner:

- We cleaned and prepared the data
- Applied machine learning models for classification
- Validated models and interpreted performance
- Applied text mining to extract qualitative insights

These techniques can be used by businesses to make better, data-driven decisions and improve overall efficiency.

